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# Flame imaging as a diagnostic tool for industrial combustion

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**Abstract**

Different approaches for condition monitoring of combustion systems based on the capture and processing of flame images are presented and discussed. The objective is to devise methods capable of converting geometrical and luminous data into reliable information on the state of practical combustion systems. Suitable processing methods are needed to extract representative information from flame images, as they contain large amounts of data whose physical interpretation is not, in general, straightforward. A thorough experimental programme has been conducted in a model industrial burner, in order to create a database of flame images for subsequent analysis. One of the options is the extraction and analysis of representative image features; even though the parametric studies revealed good sensitivity to changes in combustion regimes, this approach may entail some information loss and requires adaptation to the specific system. The other two methods explored are designed as flame identification techniques based on the whole image. One of them use self-organising feature maps and yields as output the most probable combustion regime, among those previously characterised. The other one is an adaptation of a speech recognition method and informs about the probability of an unknown state to correspond with the different combustion regimes. Their performance was tested in terms of their success rate for identification as well as on their capabilities to estimate $NO_x$ emission, as a representative outcome of the combustion process. Results in other combustion situations (oil burner and premixed combustor) are also reported. The good results obtained in all cases are thought to support the potential of the methods described for flame monitoring.
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*Keywords:* Flame images; Combustion monitoring; Classification; Identification; Neural networks

## 1. Introduction

Visualisation methods have been historically an invaluable diagnostic tool in fluid mechanics and combustion, since information on the spatial distribution of the relevant variables (even just as qualitative patterns) can be most helpful to describe, or even understand, important features of a flow or a flame. A wide range of imaging techniques are nowadays available, enabling qualitative and/or quantitative descriptions of many different variables (see, e.g., [1]). In principle, many of them might be exploited to characterise industrial flames by providing direct information from the combusting flow, which is recognised to be a key step in order to develop cleaner and more ef-

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ficient combustion systems [2,3]. However, practical reasons still prevent the use of many of them: lack of ruggedness, high cost, difficult optical access, etc. In the current state-of-art, the recording of radiation naturally emitted by the flame appears to be a feasible method; in particular, it avoids the need for seeding or external illumination and is amenable to low-cost, rugged CCD cameras. This kind of methods has been explored in a number of works, and the results are thought to confirm its suitability for practical applications.

Different approaches are possible. Some methods are oriented to derive quantitative information for some combustion variables. Pyrometry can be used to obtain temperature distributions over large regions of the flame; two-colour techniques (either in the infrared or simply using two channels of a conventional RGB camera) are the most common choice due to its reduced sensitivity to uncertainties in emissivity or attenuation. Refs. [4–6] describe some application examples in coal flames. Emission spectroscopy requires collecting radiation in narrow bands associated to chemiluminescence of excited radicals (e.g., $OH^*$, $CH^*$), whose concentration can be related to the location of reaction fronts or to the magnitude of heat release rates [1,2]. Hence, the emission at specific wavelengths might be more closely related to the combustion state than wide-band radiation, and some of the works oriented to the diagnostic/control of industrial flames [7–10] are based on the analysis of bandpass-filtered images.

Alternatively, flame images can be interpreted as a 'signature' of a particular combustion state. This approach has been applied in several works, always aimed at developing novel monitoring techniques (and, ultimately, advanced control strategies) for practical applications. Instrumentation requirements are relatively little demanding: conventional CCD cameras and frame-grabbers are, in principle, enough to collect the data needed; most of the difficulties with the hardware are associated with installation in the combustion chamber and protection of instruments from high temperatures and fouling. The main challenges are thought to be related with data analysis and, above all, the conversion of raw images into usable information.

The most common approach is 'feature extraction,' i.e., the derivation of significant parameters from a flame image. In the first place, such a procedure greatly reduces the amount of information needed to represent an image. In the second place, some features might be representative of particular behaviours of practical interest, due to their intrinsic meaning or their relationship with certain combustion conditions (e.g., flame length can be relevant for some applications or, in some cases, might be used as an indicator of staged combustion). The features analysed consist of geometrical parameters [10–12], different variables related to the level and spatial distribution of luminosity [13,14] or colour [15–17], as well as combinations of several of those properties [6,18]. Characteristic frequencies have also been derived from high-speed image records, as a relevant feature of the process either by itself or combined with other image properties [6,14,19].

The existence of relationships among combustion conditions and selected image features has been demonstrated by parametric studies reported in a number of works [6,12,13,18–20]. But, in most cases, those parameters do not have an intrinsic physical meaning, and serve mainly as indicators of a particular combustion state. Therefore, any attempt to exploit flame images for practical purposes involves as a necessary subsequent task that of relating those variables with known situations or with real combustion parameters. A wide range of approaches and processing methods have been reported. Kurihara et al. [11] proposed an empirical index for the prediction of $NO_x$ emissions as a function of some selected geometrical parameters. Shimoda et al. [4] applied a highly simplified physical model to relate unburnt carbon in a coal fired plant with luminosity features derived from flame images collected at two wavelengths. Baek et al. [15] found a linear relationship between a chromatic parameter calculated from flame images and the emissions of $NO_x$ and unburnt carbon in a pilot coal-fired furnace. Rahman et al. [14] proposed two 'quality indices,' combining parameters related to luminosity and fluctuation frequency derived from flame images, as performance indicators for utility boilers. Yu and MacGregor [17] used partial least squares to predict $NO_x$ and $SO_2$ emissions and heat losses in an industrial boiler from a set of image features (related to its geometry, luminosity and colour).

Artificial neural networks (ANN, thereafter) have been used as fitting tools to relate image features with relevant combustion parameters. Lu et al. [10] used a feed-forward ANN to predict successfully the combustion regime from 5 geometrical/luminous parameters of the flame. ANN and fuzzy logic were combined in [8,21] to adjust air and fuel flow rates, based on flame brightness and length. Wang et al. [22] applied a back-propagation ANN to relate a set of image parameters (average and deviation of temperature, ignition distance) with $NO_x$ emissions. Bae et al. [23] used a neural network to distinguish flame on/off conditions from the analysis of luminosity distributions.

A different, relatively little explored approach consists in treating the image as a data set that can be processed as a whole in order to 'classify' or 'identify' the image as representative of a particular combustion state. Allen et al. [9] reported notably

good results using artificial neural networks to classify different combustion states using the full image as input data.

Finally, the use of flame images for control or optimisation purposes has been examined in a few works (all in laboratory rigs). Allen et al. [9] and Lu et al. [10] coupled an ANN-based state predictor and a closed-loop control to bring the system to the desired state. Burkhardt et al. [8,21] combined ANN and a fuzzy controller to reach optimal conditions. Soot formation was avoided in [16] using a fuzzy controller whose inputs were the levels of blue and orange in flame images. It should be noted that all control applications mentioned require a 'state identification' stage, before a control action can be defined.

This review demonstrates the growing interest in the use of flame images for online monitoring as well as the broad range of possible methods. The approach selected and investigated in this work is explained in the next section, in the framework of a brief discussion on the possibilities and limitations of the different alternatives.

## 2. Monitoring based on flame images: Needs and approaches

A first issue in the development of any monitoring technique is to define the amount and type of information needed to characterise a process state. This might not be problematic when a system can be fully characterised by a few parameters, which is not the case for industrial flames. Their thorough description would require determining the spatial and temporal variations of many physico-chemical variables, still challenging at laboratory scale and unthinkable for the moment in real plants. Therefore, a selection needs to be made on the information to be collected. Gas composition is the type of data most widely used to monitor combustion systems, but this is a highly incomplete description of the process. A reliable diagnostic is thought to be only possible if based on some description of the flame itself.

In this context, imaging techniques appear as a most suitable alternative: they can provide data obtained directly from the flame, in the form of a 2-D distribution (still a simplification of reality, but much richer than global values). Among the different options available, a passive technique based on the collection of radiation spontaneously emitted by flames has been used in this work. This option limits the possibilities for quantitative determination of specific variables, but nowadays it is considered the most feasible alternative for practical applications. Nevertheless, it should be noted that the approaches studied in this work should be equally valid for the processing of data generated by sophisticated laser-based techniques.

A second issue is the type of approach selected for the processing and interpretation of image data. Two main categories can be established: methods oriented to the derivation and analysis of physically-meaningful parameters and those based on using sensorial information as a signature of a process state. In principle, the first one clearly appears as the most powerful and desirable. However, their development and application is not considered obvious at all. Let's consider the possibilities offered by passive imaging according to this strategy. As already mentioned, spontaneously-emitted radiation can be used to estimate some physico-chemical parameters of a flame. One of them is the determination of temperature distribution by pyrometry (actually, only valid for particle-laden flames, as Plank's law does not apply in blue flames). Another one is the use of emission spectroscopy to derive information on excited radicals from radiation collected at specific wavelengths. This kind of information, by itself, can be most valuable if the objective is the 'characterisation' of flames; however, how to exploit it for online 'monitoring' of an industrial burner is not so straightforward. This would require, for example, establishing the desired spatial distribution of $OH^*$ or the ability to interpret a particular chemiluminescence image in practical terms, and both are rarely possible in practice. Alternatively, this kind of information can be used to estimate the stoichiometry of a flame. In fact, methods based on chemiluminescence of excited radicals are probably among the most mature technologies for online monitoring in practical premixed combustors (e.g., [24–26]). However, it should be noted that even for this application, which has been analysed in many works and has a sound physical basis, no universal relationships are available and the technique needs calibration for specific fuels and facilities [25–27].

All this argumentation points to the difficulty of using directly physico-chemical parameters for online 'monitoring'; instead, empirical knowledge specifically adapted to the particular application needs to be added for converting measured data into representative parameters and/or transforming these into information adequate to monitor or optimise industrial flames. In fact, previous works oriented to monitoring from filtered images (which might be related to some physical variables, like temperature or excited radicals), actually treated the results as a characteristic pattern (*signature*) of the different flame states, instead of evaluating directly the implications of the temperature or radicals distributions regarding the 'quality' of the flame. Only in a few cases this was attempted, by means of simplified models, specific of particular applications [4,7]. In general, the

methods (with the exception of [9]) are thought to require careful design and adaptation in order to select the most representative parameters and to develop the empirical rules needed for flame monitoring in a new application. None of the works reviewed reports implementation in different combustion situations.

The treatment of image data as a 'signature' of a flame state is an alternative that can be compatible with any type of diagnostic technique: filtered spontaneous radiation (pyrometry, chemiluminescence), laser-based techniques (e.g., PLIF), broadband radiation (explored in this work), etc. In this case, the basic assumption is that, due to the huge amount of information contained in an image (intensity/colour for each pixel), the probability that different combustion states produce the same set of data is very low. As a fundamental advantage, this type of methods does not require interpreting the highly complex physicochemical phenomena involved. At the same time, its empirical nature can also be an important weakness. Whether this is a serious problem or not will largely depend on how the 'empirical knowledge' is developed and applied. For a method to become a feasible monitoring technique, it should be of a generalisable nature in the sense that implementation in new applications does not require a costly design/adaptation procedure and includes as few as possible *ad-hoc* models or parameters.

The aim of this paper is to evaluate different image processing methods regarding their capabilities for the identification of combustion states. Three different approaches have been applied (feature extraction, ANN-based identification, probabilistic pattern recognition), treating images from a conventional CCD camera as a signature of flame states. The aim was not to develop and fine-tune a method specifically adapted to a particular situation, which would be of limited interest, but to assess the potential of the approach for a relatively general case. In order to approach this objective, (1) all results presented were obtained by applying the methods as initially conceived, without any *ad-hoc* work (apart from some details regarding pre-processing of raw images), (2) the methods were applied to different burners, again with no *ad-hoc* adaptation (for brevity, only the results obtained with an air-staged burner are described in detail) and (3) the use of knowledge or rules specific to the particular burners used was avoided.

## 3. Experimental facility

The tests have been performed in a water-cooled experimental furnace, equipped with an industry-type natural gas burner (Fig. 1). The burner includes a number of adjustable settings, enabling the study of a wide range of flame types. Combustion air enters through two concentric air injections, each with a variable swirl generator (moveable blocks). A broad range of conditions, from conventional, fast-mixing flames to low-$NO_x$, air-staged combustion, can be achieved by means of a proper adjustment of air distribution and the two swirl settings. Further details on this facility and a detailed characterisation of the different combustion regimes can be found in [28].

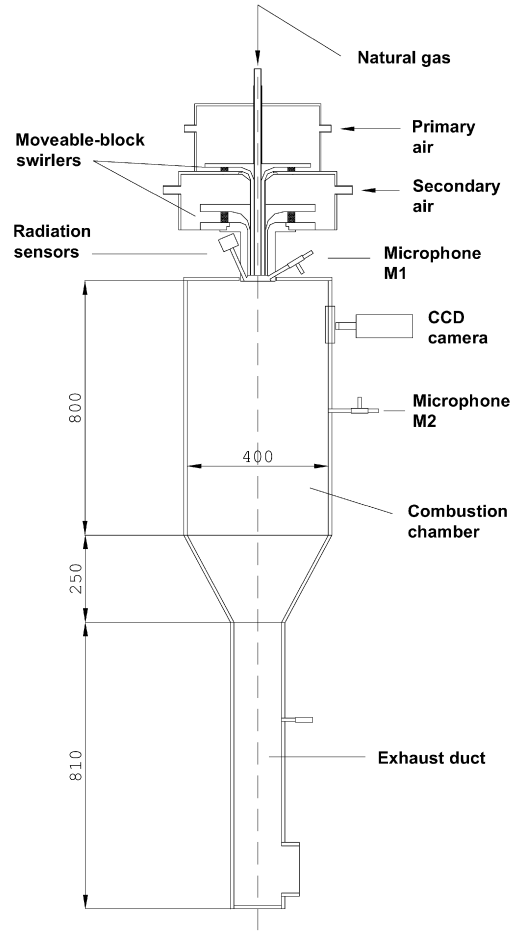


Fig. 1. Test facility and instrumentation.

The flow rates of fuel, primary and secondary air were measured with thermal mass-flow meters and controlled by closed loops acting on the respective regulation valves, whereas swirl numbers were adjusted manually. The mass-flow rate of natural gas was fixed to 5 $m^3$ N/h throughout the tests and the equivalence ratio was kept at 0.88 (3% $O_2$ in flue gases, by vol. db). Gas sampling was performed using a specific probe located at the exit of the combustion chamber, connected to a bank of individual on-line analysers for $O_2$ (paramagnetic sensor), CO, $CO_2$ (NDIR) and $NO_x$ (chemiluminescence).

Three of the burner settings were varied in the experiments: air distribution between primary and secondary registers (air ratio, AR, expressed as fraction of secondary air over the total air flow rate) and the swirl numbers of both air streams (S1 and S2, respectively; both reported here as the % with respect to maximum swirl level). The different combinations of S1, S2 and AR tested led to a wide range of flame patterns. $NO_x$ emissions ranged from 15 ppm for deep air-staging conditions (AR = 100%) up to 44 ppm for fast-mixing flames. Detailed in-flame measurements revealed significant changes in the distribution of species and temperatures inside the flame when AR, S1 or S2 were varied. The dynamic behaviour of the flames was analysed from time-resolved measurements of pressure and OH* chemiluminescence (radiation at 310 nm), yielding widely different results: some flames were highly stable (AR < 60% and high swirl levels), others displayed significant fluctuations but pressure and radiation signals were uncorrelated (AR = 60–80% or low swirl levels), whereas high values of AR led to strong correlations of both signals as a clear indication of the onset of thermo-acoustic instabilities. The experimental characterisation of the different combustion regimes has been reported in [28]. No detailed description of the flames is included here; firstly, for brevity and, secondly, because it is not essential for the purpose of this work. Exploiting the knowledge gained from detailed measurements for flame monitoring from images was avoided on purpose; whereas this might add 'some physics' and even lead to improved performance, the procedures so developed would lose much of their generality. Hence, the monitoring methods presented here do not use any qualitative or quantitative piece of information not contained in the very flame images.

Flame images were captured using a CCD camera (Panasonic WV-CL350) housed inside a water-cooled jacket, installed in the furnace wall and oriented perpendicularly to the burner axis, so that a full picture of the flame can be obtained. The use of a pinhole lens together with a purge of compressed air maintained the optics clean and cool. Monochrome images (8-bit greyscale, 320 × 240 pixels, exposure time 1/60 s) were recorded in a process computer through a frame-grabber board (Data Translation DT3155). Image records were collected for a total of 49 different flames, corresponding to different combinations of S1, S2 and AR.

Since objective is the exploration of feasible methods for the monitoring of practical flames, an essential feature of this study is the use of conventional video cameras. This imposes some limitations on the information that may be recorded. The radiation collected by the CCD corresponds basically to the visible range; for blue flames, this includes chemiluminescent emission from different radicals: narrow bands for CH*, $C_2^*$, broadband radiation from $CO_2^*$ and other minor contributions [29]. The camera performs a line-of-sight integration, so that each pixel accumulates contributions from points at different distances. As explained below, representative images were obtained by averaging several frames; hence, fluctuations due to turbulence or (much more important in some cases) flame oscillations are smoothed out. Therefore, although there must be some connection between the images recorded and the physics of the flame, the lack of resolution in wavelength, space and time makes the derivation of some kind of combustion variable from them very difficult. Temporal or spatial averaging can be a serious drawback for monitoring methods based on physical parameters (e.g., derived by pyrometry or chemiluminescence), but they are not necessarily when the image is treated as a signature. By no means does this imply that low resolution is a positive feature. On the contrary, a better resolution in wavelength, space or time might produce images more representative of the process; but this would require much more expensive and delicate instrumentation (e.g., intensified CCD, filters, several cameras for 3-D reconstruction, etc.). Furthermore, whereas a wide range of conventional, inexpensive CCD cameras are nowadays available that might provide images with better definition, the good results reported with a monochrome, 340 × 240, 8-bit images serves to demonstrate the low requirements in this respect of the methods studied.

## 4. Feature extraction

Image analysis algorithms were used to derive a wide range of parameters from the frames recorded at each of the operating conditions tested. An image was represented by two different functions (see Fig. 2):

- *Intensity function*: $I(x, y) = 0$–$255$, where $x$ and $y$ are pixel coordinates.
- *Presence function*: $I^*(x, y) = 0$ when $I(x, y) = 0$ and $= 1$ when $I(x, y) > 0$.

The presence function was then used to calculate various geometrical parameters: area, width distribution, maximum width, length, centre of geometry and angle at the flame root. Features related to luminosity distribution were derived from the intensity function: centre of mass, discrete moments, and a number of statistical parameters derived from the intensity histogram, such as mean, variance, skewness or kurtosis. Fig. 2 illustrates how some of those parameters are derived from a flame image.

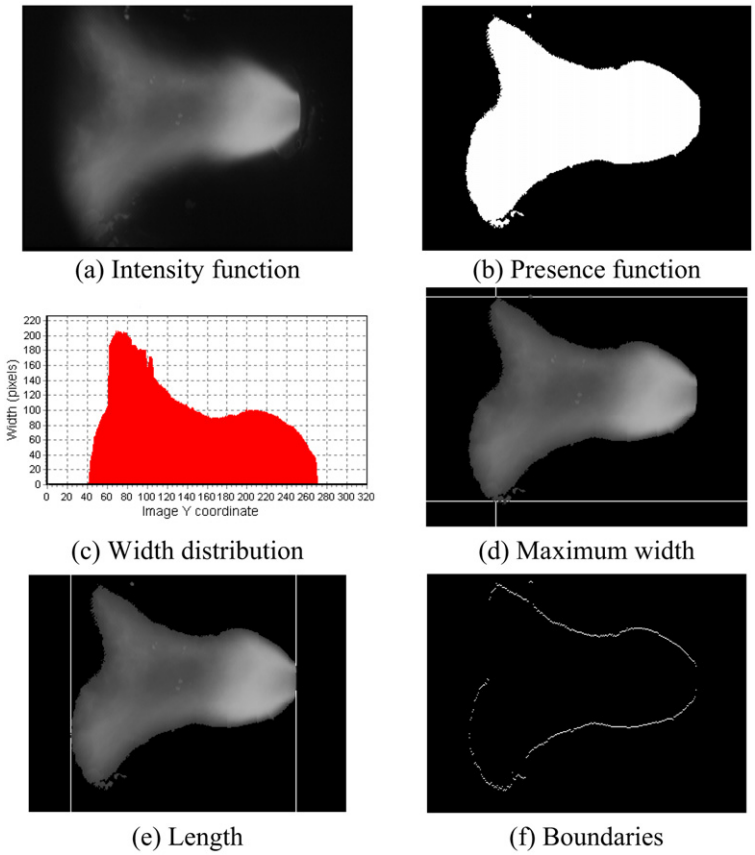


Fig. 2. Extraction of some features from a flame image.

The next step was to explore the sensitivity of the different parameters to variations in operating conditions. For this method to be valid, modifications in burner settings must cause a meaningful change (exceeding its experimental uncertainty) in a particular image feature. For brevity, only a few examples are shown in Fig. 3 for variations in the swirl levels of the two air streams. The intervals drawn at each data point represent the dispersion about the mean value, calculated over the full sequence of images recorded in a test. Increasing swirl numbers led, as expected, to more compact flames with smaller area and length.

The final objective is to identify the combustion state from selected features of the flame images; this requires some sort of quantitative relationship among burner settings and image features. With this purpose, multiple linear regression was applied to the full set of experimental results. The binary correlation coefficients were below $\pm 0.5$ in all cases, indicating weak linear relationships. As shown in Fig. 3, even though some variables seem to display well-defined dependences, their non-linearity justifies the low correlations obtained. Therefore, other types of functional forms, specifically adapted to each feature, should be used to represent adequately the relationships observed. This is thought to be a significant shortcoming if the objective is to develop generalisable diagnostic methods.

Some trials were also performed using artificial neural networks as the fitting algorithm, due to their ability to handle multivariate, non-linear relationships. Although this method provided reasonable results with some combinations of image features, the capabilities for state identification were poorer than with the approaches explained in the following sections. The cause is thought to be the information loss associated to the replacement of the full image by a few representative parameters.

## 5. Identification of combustion states with ANNs

Artificial neural networks (ANN) are mathematical algorithms inspired on the structure of biological neural systems and the way information is processed by the brain, with good capabilities to handle complex information with unknown and highly non-linear functional relationships among the different variables.

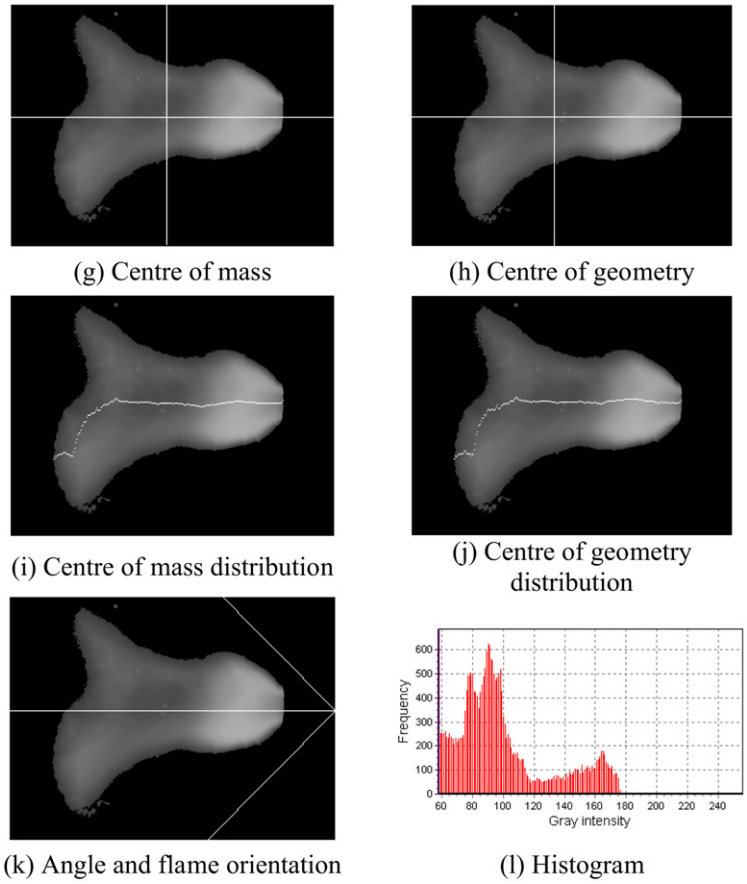


Fig. 2. (*continued*)

Ref. [30] describes fundamental and application aspects of these methods.

Images are an outstanding case of complex data sets, whose processing and/or interpretation has motivated great efforts during the last decades. As described in [31], ANN-based algorithms can be very effective for most of the tasks usually faced in image processing. Object recognition based on features is one of them; in fact, ANNs have been successfully used in a number of works specifically oriented to identify combustion states (or, equivalently, to estimate performance parameters like pollutant or unburnt emissions) using as inputs some selected features previously extracted [8,10,22,23]. As noted at the end of previous section, this procedure was also tested in this work, although its performance was not as good as with the other methods reported.

Alternatively, all pixels forming an image can be directly used as input data (i.e., the number of representative features equals the number of pixels). The main drawback is the large amount of data to be processed; the advantage is that information loss due to data reduction is avoided. The only experience found in the open literature is the work by Allen et al. [9] where combustion states were successfully identified using a back-propagation ANN fed with the intensity levels in 1024 pixels.

As noted in [31], self-organising feature maps (SOFM, thereafter) are suitable for object recognition based on pixel data, and was the method selected in this work. The so-called Kohonen maps are used here, which have been widely applied for classification purposes, especially in pattern recognition [32, 33]. A SOFM consists of neurons organised on a regular low-dimensional grid. The neurons are connected to adjacent ones by a neighbourhood relation dictated by the topology of the map (hexagonal in this case). Each neuron in the map contains a weight vector with the same dimension than the input vectors and represents a ‘pattern’ flame image. During training the input vectors are compared with that of each neuron by calculating a distance (Euclidean in this case) and the weight vectors of the closest neuron (winner neuron) and its neighbours are modified according to the so-called Kohonen learning rule. This process is iterated many times so that the distances

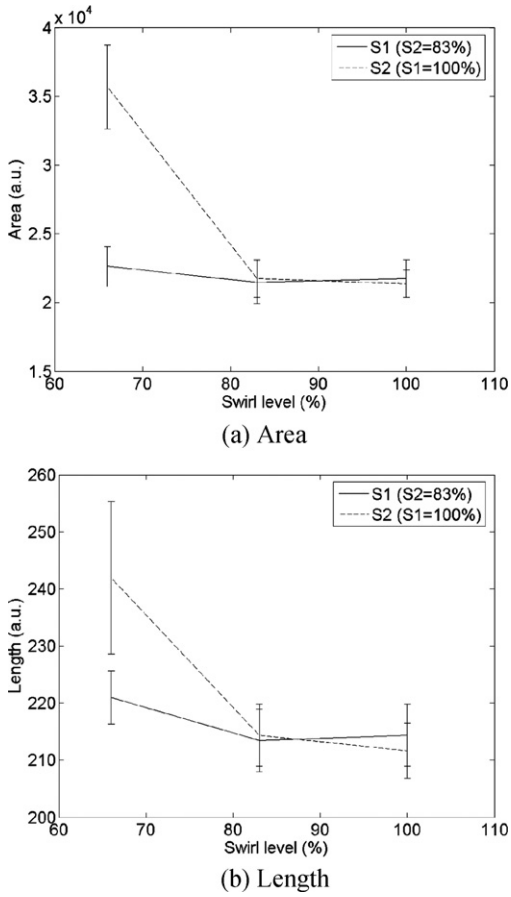


Fig. 3. Effect of swirl settings on flame area and length. AR = 75% in all cases.

between the input vectors and the winner neurons and the distances among neighbour neurons get smaller; the final results is a set of 'pattern images' that are ordered in the map according to their visual features. The interested reader can find more detailed descriptions in the references quoted. Once the SOFM has been trained, an 'unknown flame' representing an 'unknown state' is classified by the 'winner neuron' when it is shown to the SOFM; identification with respect to 'known states' consists in selecting among the flames included in the database used for training the case yielding the same winner neuron.

### 5.1. Image pre-processing

Different values of the three burner settings (S1, S2, AR) were combined to yield a total of 49 different combustion states. Sequences of 100 images (greyscale, 320 × 240 pixels) were captured for each of the tests. As a previous stage, the images were preprocessed in order to reduce the size of the input vector as much as possible without loosing information as well as to strengthen some relevant characteristics. The steps followed were:

1. *Average*: All images recorded for a single test were averaged. This step is not only very useful for data reduction (99%) but also to filter out fluctuations among different frames, resulting in a more representative image of the flame. This was ensured by averaging over 100 images, although most probably a much lower number would suffice.
2. *Contrast*: Luminosity values were normalised, so that the darkest pixel of the image was set to 255 (white in greyscale) and the brightest to 0 (black). This operation should not be interpreted as an information loss because absolute luminosity depends on many factors (CCD calibration, lens, attenuation along the optical path, etc.) and its consideration could make the procedure highly dependent on the particular setup.
3. *Segmentation* (thresholding): The flame area was delimited by setting to 0 all pixels with intensity $<60$.
4. *Trimming*: After completing the previous operations, large areas of the image consisted of black pixels. In order to discard useless data, the flame images were trimmed by a rectangle built from the disjunction of the smallest rectangles enclosing non-black parts of the individual images. Such rectangular area was defined at the training stage, and also used afterwards to trim any new image. As a result, images were reduced from 320 × 240 to 170 × 200 pixels.

### 5.2. Configuration and training of the SOFM

Initially, the topology and number of cells of the SOFM need to be defined. For this application, a 3-by-10 map might be suitable to represent the matrix of test cases: 3 adjustable burner settings and a maximum of 10 different values for each of them. However, a larger map was preferred to favour a clear topological isolation among different groups of flames, and a 10-by-10 map was finally used.

As a first step, the ANN was trained with images captured in 49 different combustion regimes. The SOFM was trained for 1000 epochs. In each of these epochs, the distance between a particular flame image and every neuron in the SOFM (defined by a set of weights) was calculated. The neuron yielding the shortest distance was marked as the winner and the weights of all neurons in the neighbourhood of the winner were updated using the Kohonen rule. As a result of the training process, each of the 100 neurons ends with an associated 'pattern' flame image.

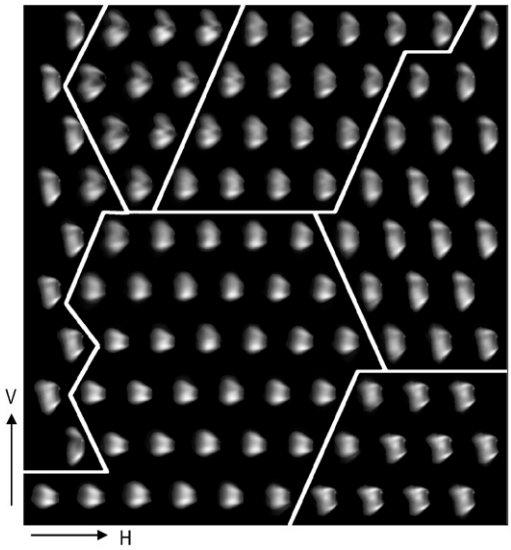


Fig. 4. Pattern flame images associated to the SOFM neurons.

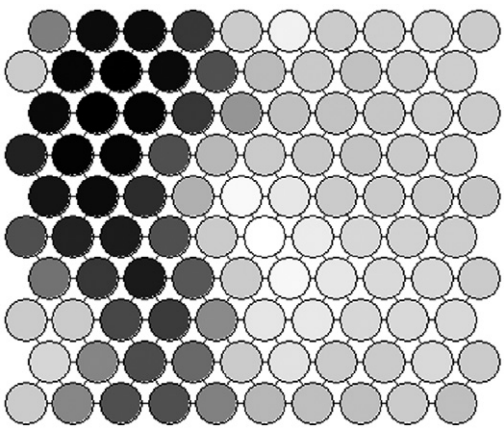

Fig. 5. Swirl level in primary stream, S1, associated to the neurons of the SOFM (greyscale: white for S1 = 100%, and black for 0%).

### 5.3. Topological classification

Fig. 4 displays the pattern images associated to the different neurons in a 10-by-10 map. This representation serves also to illustrate the way in which the SOFM operates: images are grouped according to their degree of similitude. It should be noted that although the neurons are represented equidistant in the network, actually they are not. The white lines have been drawn on the map to indicate the longest distances between neighbour neurons. Each subgroup so obtained includes images with evident similarities among them and is characterised by patterns that are appreciably different from those of other groups.

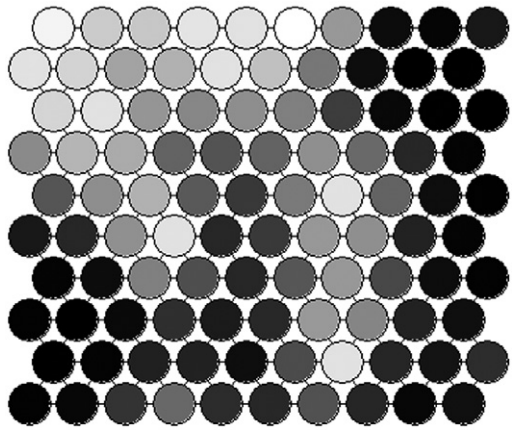

Fig. 6. Swirl level in secondary stream, S2, associated to each of the neurons of the SOFM (greyscale: white for S2 = 100%, and black for 0%).

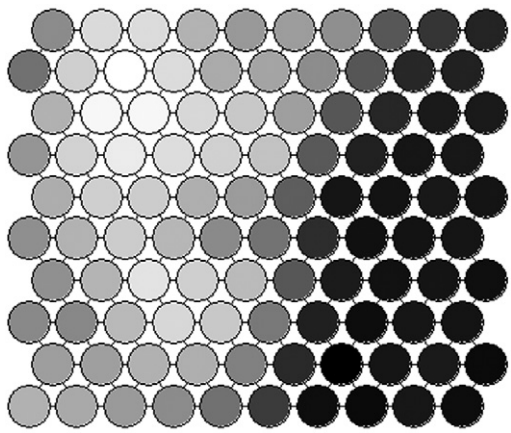

Fig. 7. $NO_x$ emissions associated to each of the neurons of the SOFM (greyscale: white indicates the highest value, 44 ppm, and black the lowest, 14 ppm).

A basic assumption behind the use of images for the diagnostic of combustion states is that geometrical/luminous properties maintain some relationship with operating conditions or system performance. The type of classification provided by Kohonen maps can be particularly useful to evaluate this hypothesis. With this purpose, numerical values were assigned to each of the 100 neurons, according to the following procedure. The parameters (burner settings, emissions) characterising each of the 49 combustion tests were transferred to the corresponding winner neuron. An interpolation procedure was applied to assign numerical values to those neurons without an associated 'real' combustion regime, using the two coordinates, $V$ and $H$, defining its location in the map (Fig. 4) as independent variables. Figs. 5–7 display the re-

Table 1
Measured $NO_x$ emission for the neurons in the bottom-left zone of Fig. 4 with an associated real flame

| Neuron coordinates | | Measured $NO_x$ (ppm) |
|---|---|---|
| $V$ | $H$ | |
| 1 | 1 | 37.6 |
| 1 | 1 | 37.9 |
| 1 | 1 | 38.2 |
| 1 | 3 | 35.8 |
| 1 | 4 | 34.4 |
| 2 | 5 | 33.4 |
| 3 | 1 | 41.5 |
| 3 | 2 | 35.8 |
| 4 | 3 | 22.7 |
| 4 | 5 | 22.8 |

sults obtained in terms of the swirl levels in the two air streams and $NO_x$ emissions, respectively. The following example describes how this assignation procedure was implemented. The location of the neurons in Figs. 4–7 will be described with the coordinates $V$ and $H$ (both in the range 1–10). Let's take the image at $V = 2$, $H = 4$ in Fig. 4; like the other images shown in Fig. 4, it does not correspond to a real flame but was created internally by the SOFM algorithm from the set of 49 images used for training. In a subsequent stage, the SOFM was used to assign a neuron to each of the 49 flames. Table 1 displays the results: some of the neurons were associated to a real flame image (or, in a few cases, to several, like neuron $V = 1$, $H = 1$), but others were not (e.g., $V = 2$, $H = 4$). Those with an associated real flame were assigned its measured $NO_x$ emission. For the other neurons, a $NO_x$ value was calculated by linear interpolation from actual measured emissions assigned to neighbour neurons, using the coordinates $V$ and $H$ as independent variables. As a result, a $NO_x$ concentration of 37.6 ppm was calculated for the neuron at $V = 2$, $H = 4$; this method enabled assigning a $NO_x$ value to all 100 neurons as shown in Fig. 7. An equivalent procedure was applied to build Figs. 5 and 6 for S1 and S2, respectively.

As shown in Fig. 5, the different degrees of swirl in the primary air become automatically grouped in the SOFM. Since other burner settings (S2 and AR) were also varied, flames with the same S1 can display widely different aspects, which explain that similar S1 values can be found in different regions of Fig. 5. Similar comments are applicable to Fig. 6. This 'automatic' grouping of cases with similar burner settings is, obviously, due to the fact that precisely those settings determine the aerodynamics and mixing pattern in the flame, and therefore its visual appearance.

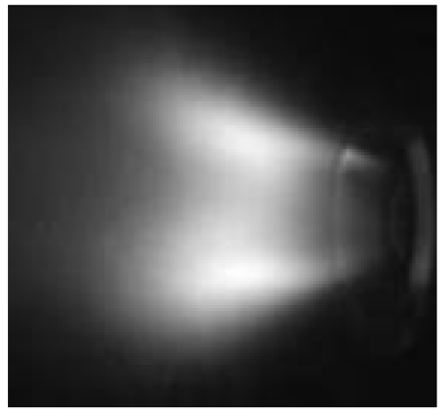

Fig. 8. Average flame image recorded with the burner adjusted at AR = 50%, S1 = 100%, S2 = 50%; measured $NO_x$ emission was 37.7 ppm.

### 5.4. Estimation of $NO_x$ emissions using flame images

An alternative viewpoint is to evaluate whether images and $NO_x$ emissions (two different results of the combustion process) display some correlation. This can be observed in Fig. 7, displaying the $NO_x$ emissions assigned to the 100 neurons of the SOFM. The different emission levels are noticeably grouped over the map, with gradual transitions between one neuron and its neighbours. Since this distribution is directly obtained from the distribution generated by the Kohonen map according to image properties, the close relationship among both aspects becomes apparent.

Such strong correlation might be exploited to infer the $NO_x$ emission of a particular flame using the image as the only input. With that purpose, a set of 42 new image records (different from those used for training) were classified using the SOFM developed; i.e., each of them was assigned to one of the neurons in the 10-by-10 map. The $NO_x$ emission estimated for that flame was the value previously assigned to its winner neuron, as shown in Fig. 7. For example, the flame image shown in Fig. 8 was recorded for AR = 50%, S1 = 100%, S2 = 50%, and the measured $NO_x$ emission was 37.7 ppm. Classification with SOFM yielded as winner the neuron at $V = 2$, $H = 4$. As explained above, no measured $NO_x$ was available for this neuron, and the value assigned by interpolation was 37.6 ppm. Therefore this was the value of $NO_x$ predicted from the image in this case, very close to the actual value of 37.7 ppm. Actual $NO_x$ emissions and estimations with this method are compared in Fig. 9. A fairly good agreement is obtained, with deviations <3 ppm in most cases; in particular, high- and low-$NO_x$ regimes are perfectly distinguished.

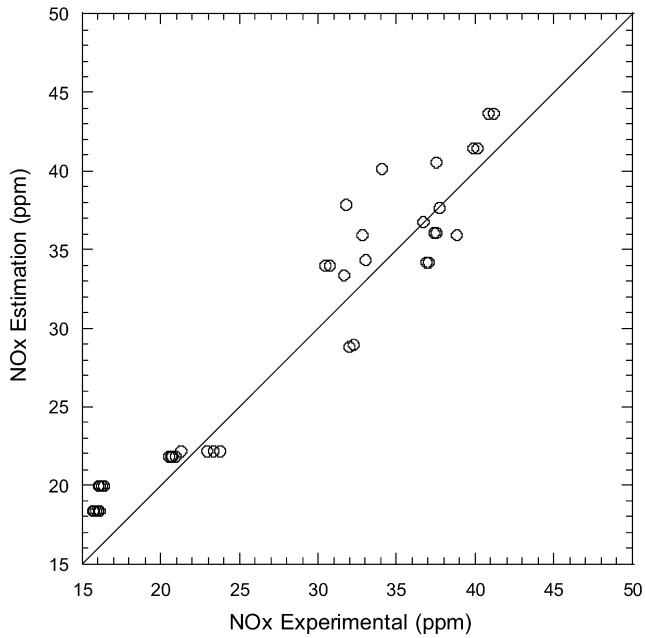


Fig. 9. Comparison between predicted and measured $NO_x$ emissions. Predictions based on classification with SOFM.

This is thought to be a most relevant result for practical applications, as the good correlation obtained among images and $NO_x$ emissions might enable the development of advanced diagnostic methods by using flame images to estimate $NO_x$ emissions (i.e., a 'virtual' $NO_x$ sensor). This could open new possibilities for the monitoring of emissions from individual flames in multi-burner furnaces, or to greatly reduce the response time of gas analysers down to the levels acceptable for control purposes.

Alternatively, the Kohonen map can just be used to identify the combustion state with respect to others previously known (used at the training stage). The process of assigning a winner neuron to a new input image yields a particular location in the map that can coincide with a particular combustion regime, or can be used to find the most similar regimes as those located at short distances. The comparison between estimated and actual $NO_x$ emissions serves also as an indication of the good accuracy of the SOFM as a flame identification tool.

## 6. A probabilistic classification method

The result of most identification methods (e.g., SOFM) is the state, among those previously known, that most probably corresponds to the unknown state being processed. A shortcoming is that the reliability of this result is not qualified. It would be most desirable, for example, to obtain some information about the degree of coincidence with respect to the state assigned or, if the unknown flame displays some similarity with several states, the relative probabilities associated to each of them. This would enable, among others, the detection of off-design operations, evaluating how reliable is the identification or combining the result with other sources of information.

A different approach has been applied for the classification of flame images that provides information on the probability of coincidence with each of the combustion states previously known. This procedure is inspired on the cepstral analysis techniques, commonly used for speech recognition [34]. Although sound records are of a type different from image data, both can be equally transformed into covariance matrices, as required in this kind of methods.

### 6.1. Image pre-processing

Ten black and white images (320 × 240 pixels each) were recorded for each of the combustion regimes tested. Each of them is obtained by averaging ten individual frames in the sequence captured.

As a first step, Principal Component Analysis (PCA, thereafter) was applied for data reduction. Given the large amount of data needed to represent the images, minimising the size of the input vectors entails significant benefits, especially when real-time performance is required. PCA is a convenient method for image processing, because it allows reducing the dimension of the input space while maintaining much

of the original structure of the input images. It has been used extensively in pattern recognition [35] and has shown to be an effective method for the analysis of flame images [17,20].

Images are represented by a matrix $I$ of dimension $320 \times 240$. Previous to PCA, $I$ is transformed by rearranging image blocks into columns. First, $I$ is divided into $p$ square blocks (if necessary, padded with zeros) of $s \times s$ pixels; second, a new matrix $X$ with $s^2$ rows and $p$ columns is formed by filling each column with the values in one of the blocks. Finally, PCA is applied to derive a lower-dimensional representation. Given a matrix $X$, and the desired new dimension, $q$, the PCA algorithm yields the matrix $M$ ($q$ rows and $s^2$ columns) and the vector $B$ ($q$ components) needed to obtain a matrix $Y$ ($q$ rows and $p$ columns), according to

$$Y_{ij} = \sum_{k=1}^{s^2} M_{ik} X_{kj} + B_i, \quad i = 1, \ldots, p. \tag{1}$$

$Y$, $M$ and $B$ can be taken, therefore, as an alternative representation of the original matrix $X$. A parametric study on the influence of $s$ (in the range 5–18) and $q$ (from 5 to 20) yielded as the optimal combination $s = 14$, $q = 12$. Finally, $Y$, $M$ and $B$ were put together in a new matrix $Z$, with $q$ rows and $p + s^2 + 1$ columns. Cepstral analysis techniques were applied on the matrices $Z$. The corresponding covariance matrix, Z_COV, is calculated as

$$(\text{Z_COV})_{ij} = \sum_{k=1}^{s} Z_{ik} Z_{kj}^t. \tag{2}$$

A library of covariance matrices was created by calculating Z_COV for nine of the images stored for each of the $N$ combustion regimes tested; therefore, each combustion state is characterised by $N_1 = 9$ different data sets. The degree of similarity among different images was compared by calculating a 'distance' between their respective covariance matrices. The parameter chosen was the arithmetic harmonic sphericity (AHS) distance, defined for two generic matrices $A$ and $B$ of dimension $m$ as

$$D(A, B) = \log_{10}\left(Tr\left(AB^{-1}\right)Tr\left(BA^{-1}\right)\right) - 2\log_{10} m. \tag{3}$$

The important data reduction rate achieved should be noted, since only $q^2 = 144$ data contained in Z_COV are finally used to evaluate an image, instead of 76800 values in the original matrix $I$.

### 6.2. Probability calculation and pattern selection

Classification is performed by calculating the probability of an unknown 'test flame' ($x_i$) to be like each of the $N$ 'pattern flames' in the library ($y_j$), $P(x_i \sim y_j)$. In general, each 'test flame' is characterised by different 'test images,' $x_{i,n}$ with $n = 1, \ldots, N_2$, so the probability is averaged over all of them:

$$P(x_i \sim y_j) = \frac{\sum_{n=1}^{N_2} P(x_{i,n} \sim y_j)}{N_2}. \tag{4}$$

$P(x_{i,n} \sim y_j)$ is calculated as the probability for a particular 'test image,' $x_{i,n}$, of 'test flame' $x_i$ to have a null AHS distance from the matrices characterising a 'pattern flame' $y_j$. Since each 'pattern flame' is defined by $N_1$ 'pattern images' $y_{j,m}$, $N_1$ different distances can be calculated. The average of the AHS distances between the test image ($x_{i,n}$) and each of the $N_1$ 'pattern images' was taken as the representative value:

$$D(x_{i,n}, y_j) = \frac{\sum_{m=1}^{N_1} D(x_{i,n}, y_{j,m})}{N_1}. \tag{5}$$

So, the set of $N_1$ 'pattern images' is treated as an ensemble of data sets providing a statistical representation of a specific combustion regime; this is a relevant feature of the method and provides the robustness required to handle 'noisy' data (associated, for example, to turbulent fluctuations in industrial flames). This is represented statistically by a Gaussian probability distribution, $f_{yj}(d)$, for the AHS distance: different occurrences of the same combustion regime are expected to provide an ensemble of AHS distances with ~0 mean and a variance associated to this particular regime. The width of the distribution is estimated from the standard deviation, $\sigma_{yj}$, of the $N_1(N_1 - 1)/2$ distances calculated among pairs of the $N_1$ pattern images available.

The probability of a 'test image,' $x_{i,n}$, to belong to a particular pattern, $y_j$, is estimated as $f_{yj}(d)$ for $d = D(x_{i,n}, y_j)$. This is illustrated in Fig. 10 for two hypothetical patterns $y_1$ and $y_2$; if the functions $f_{y1}(d)$ and $f_{y2}(d)$ (with respective widths $\sigma_{y1}$ and $\sigma_{y2}$) are centred respectively at $D(x_{i,n}, y_1)$ and $D(x_{i,n}, y_2)$, the probability sought is directly obtained as the intercept with the ordinates axis. This example clearly shows that not only the mean value must be considered, but also the width of the distribution. Even though the average AHS distance is shorter for pattern $y_1$, the most probable combustion regime is $y_2$. Pattern $y_1$ is represented by a very narrow distribution, and the probability decays rapidly even at short AHS distances; on the contrary, pattern $y_2$ is characterised by a wider range of covariance matrices, and the probability of actually representing an unknown state maintains significant values up to relatively high AHS distances.

Fig. 11 displays the classification tests performed for two of the flames. The test case on the left (#5)

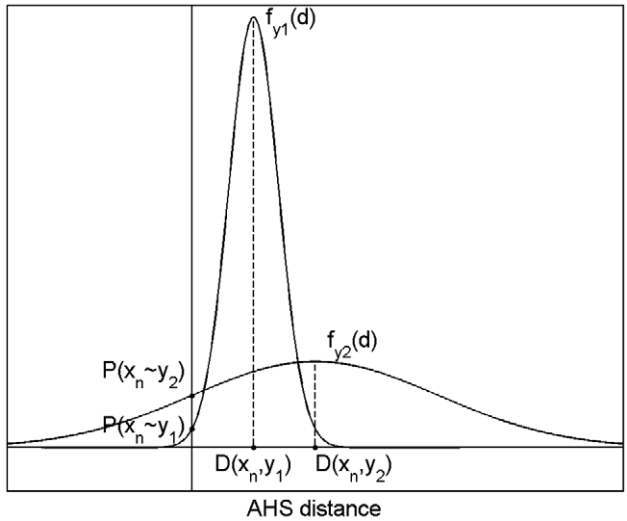


Fig. 10. Evaluation of the probability that a test flame $x_n$ corresponds with pattern flames $y_1$ and $y_2$.

corresponds to the image shown in Fig. 8, already mentioned in a previous section. The intercept of the Gaussians to the reference patterns, $f_{yj}(0)$, yields a measure of the probability of the test image to belong to that particular combustion regime. In some cases (Fig. 11), several of the reference cases give a non-negligible probability. To facilitate the interpretation, the results were normalised so that the sum for all patterns is 100%:

$$P(x_{i,n} \sim y_j) = \frac{f_{y,j}(0)}{\sum_{m=1}^{N_1} f_{y,m}(0)}. \tag{6}$$

For example, test image #5 yielded non-zero probabilities for three of the flames (see Fig. 11 and Table 2). Since the highest probability was obtained for the correct pattern, the identification was successful in this case. It should also be noted that flames #6 and #7 were also very similar, because variations in swirl number of secondary air in the range 0–50% had little influence on the aerodynamics of the flame for S1 = 100%, as evidenced by the very similar $NO_x$ emissions measured for the three flames.

### 6.3. Classification tests

This identification method was applied to the same $N = 49$ combustion regimes considered in the previous section. $N_1 = 9$ images were used to generate a set of covariance matrices for each of those 'pattern flames'; an additional image ($N_2 = 1$), recorded in the same operating conditions, was stored for each of the 49 cases, and then used as 'test images' to evaluate the performance of the method. Fig. 12 displays the calculated probabilities, $P(x_i \sim y_j)$, that a particular test image, $x_i$, corresponds to pattern $y_j$. A fully successful identification would be obtained if the probability is 0 if $i \neq j$ and 100 when $i = j$. Fig. 12 can be interpreted in two complementary ways:

- The outcome of the identification process is that 'test image' $x_i$ corresponds to the combustion regime, $y_j$, for which $P(x_i \sim y_j)$ is maximum. The results shown in Fig. 12 indicate that identification has been successful in 32 out of 49 cases. Nevertheless, it should be noted that all wrong identifications occurred among very similar flames, in most cases with probabilities for the winner pattern that are only slightly higher than those obtained for the correct one. This is an important advantage of this method, as the output not only includes the most probable pattern but also informs about similarities with other flames.
- The absolute value of the probability obtained for the winner pattern can be taken as an indication of the reliability of this result. Low probabilities can be due, as shown in Fig. 12, to the similarity between some reference cases. Otherwise, a low value could indicate, for example, that the unknown state is out of the range represented by the pattern flames, or that some deviation has occurred in the combustion equipment.

Similarly to the exercise reported with the SOFM, this identification tool might be used to estimate some relevant parameters of the combustion process. Fig. 13 displays the $NO_x$ estimations obtained by assigning the value measured for the corresponding winner patterns to each of the test cases. The coincidence is almost perfect in most cases; even in the cases of wrong identifications, the deviation is

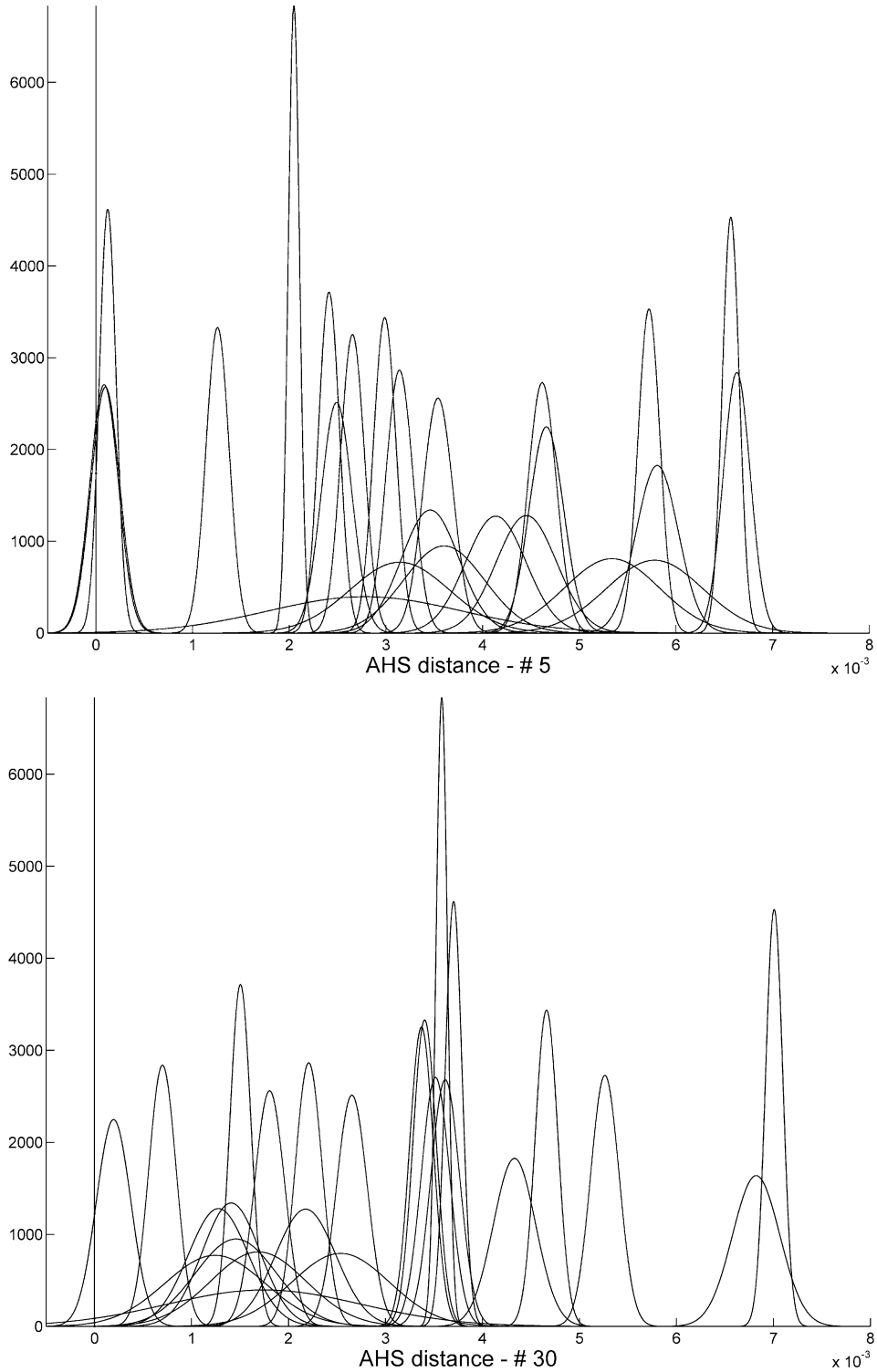


Fig. 11. Classification tests for two of the test flames (arbitrary units).

very small, further confirming the similarities between the flames with wrong or ambiguous identifications. Significant differences are only observed in the regimes with lowest $NO_x$ emissions, correspond-

Table 2
Results of the classification procedure for test image #5

| Pattern flame | AR (%) | S1 (%) | S2 (%) | $NO_x$ (ppm) | Probability (%) |
|---|---|---|---|---|---|
| #5 | 50 | 100 | 50 | 37.6 | 49.9 |
| #6 | 50 | 100 | 25 | 37.9 | 25.2 |
| #7 | 50 | 100 | 0 | 38.2 | 24.9 |

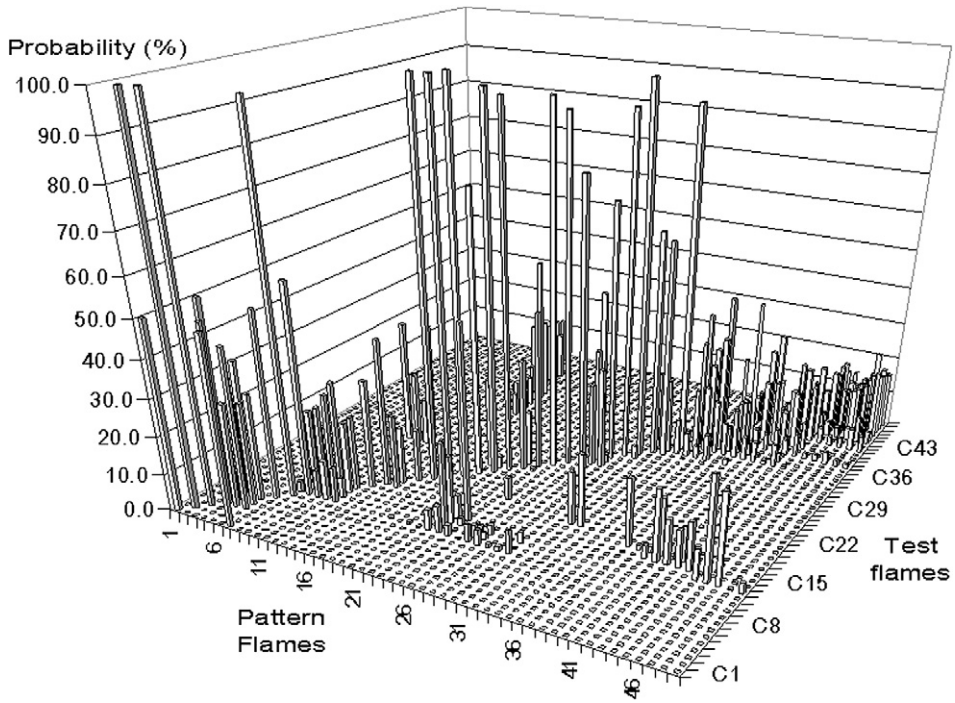


Fig. 12. $P(x_i \sim y_j)$ estimated by image identification for the 49 flames.

ing to the highest air staging ratios (AR = 90–100%). As previously noted, those flames displayed thermoacoustic oscillations, which can result in wider variability among flame images and, hence, in less certain identifications.

### 6.4. Combination with other classification criteria

As already noted, an advantage of this approach is that the result obtained is not limited to a yes/no answer, but also quantifies the degree of similarity with respect to all the patterns available. This can be further exploited, for example, to combine this type of diagnostic with other potential sources of information.

This possibility is illustrated by using $NO_x$ emission as a representative feature of the different combustion states. Similar steps were followed in order to obtain a classification based only in $NO_x$ emissions. The probability distributions associated to the 'pattern flames' consisted of a Gaussian function with the measured $NO_x$ emission as the mean value, and a standard deviation representing the experimental uncertainty ($\leqslant$2 ppm, as estimated from repeatability tests). A procedure similar to that just described was followed to estimate $P'(x_i \sim y_j)$, using the difference between average $NO_x$ emissions of the unknown and pattern cases was used instead of the mean AHS distance. The results with this procedure are presented in Fig. 14, in terms of $P'(x_i \sim y_j)$. These results demonstrate that the $NO_x$ criterion is clearly insufficient to distinguish one particular flame among all cases included in the test set. Success rate is very low and only 6 out of 49 flames yielded probabilities >20% for the correct pattern. Hence, $NO_x$ emission can help in classifying different regimes but this criterion alone is not a reliable method to identify flames.

The results obtained with the two classification methods (images and $NO_x$) can be combined together by multiplying the probability matrices. The results are normalised to 100% and shown in Fig. 15. Only slight differences can be observed between Figs. 12 and 15, due to the ambiguous predictions obtained

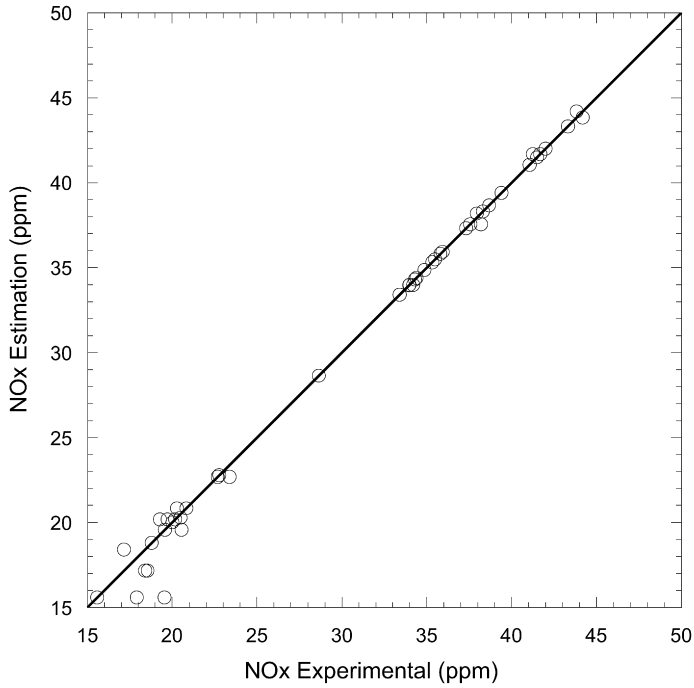


Fig. 13. Comparison between predicted and measured $NO_x$ emissions.

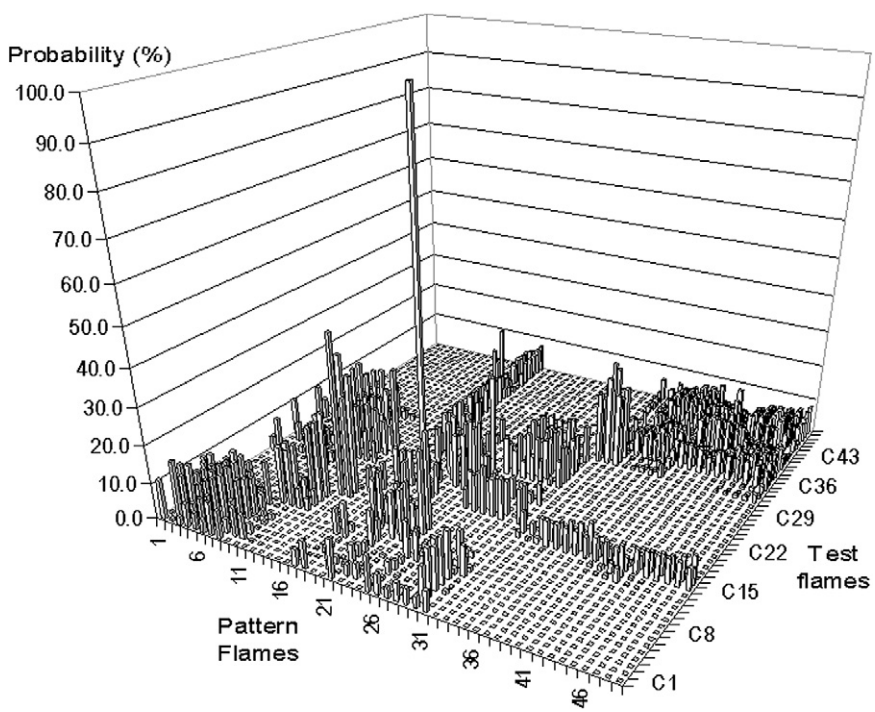


Fig. 14. $P'(x_i \sim y_j)$ estimated by comparing $NO_x$ emissions for the 49 flames.

with the $NO_x$ criterion. Nevertheless, all the changes were in the correct direction and some improvement can be seen in many of the probabilities along the diagonal. In particular, the superposition of both criteria reduced off-diagonal probabilities for test flames C11–13 and C15–17. In these and other cases, the

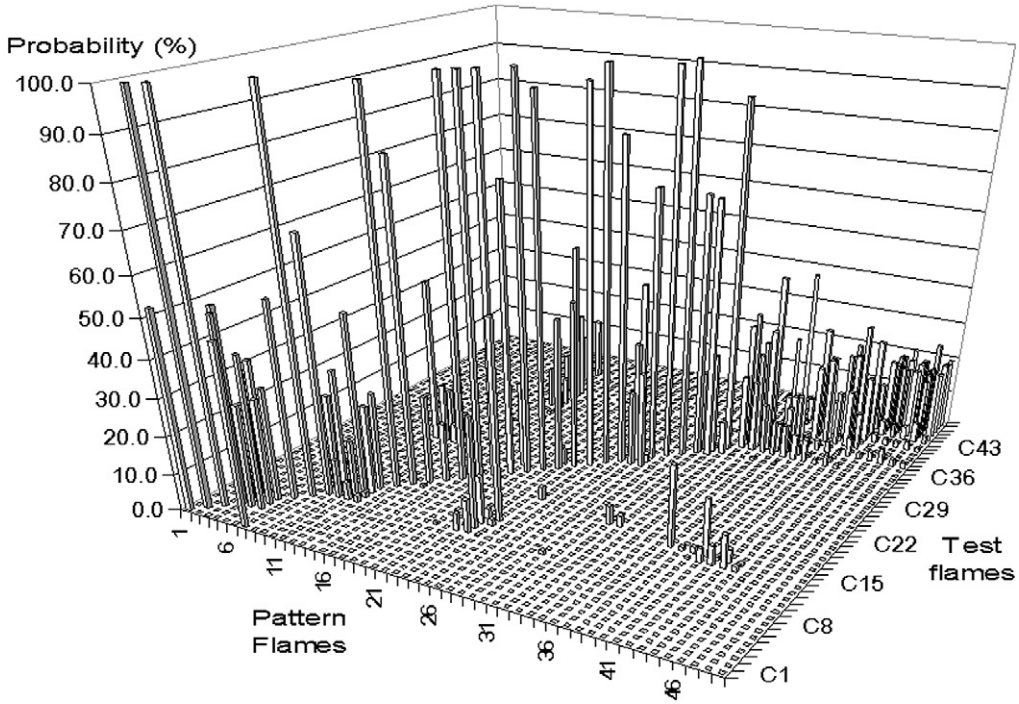


Fig. 15. $P''(x_i \sim y_j) = P(x_i \sim y_j)^* P'(x_i \sim y_j)$, for the 49 flames.

$NO_x$ criterion contributed with different weights that finally led to increase the confidence of the prediction, or even to a correct identification.

As already noted, $NO_x$ emission is not a distinctive feature among most of the flames contained in this set of cases. Nevertheless, it serves to illustrate a general procedure that could be applied using other features to distinguish among cases with similar image patterns. If that additional feature displays clear variations for the cases where image classification is ambiguous, the combined probability would help to discard the wrong pattern (if there is no feature that displays significant variations in dubious cases, probably those combustion regimes are so similar that a wrong classification does not entail any practical consequence).

## 7. Application to other combustion situations

The potential of monitoring methods should not be evaluated from the results in a particular application but strongly depends on their capabilities to handle different situations. Moreover, ideally, they should be applicable without needing a costly process of *ad-hoc* design and adjustment and should avoid as much as possible the use of specific parameters and interpretation criteria that can be adequate for one particular application but not for many others. Due to the nature of the methods reported here, it is not possible to demonstrate their generality on physical grounds; although there is a relationship among visual and physical characteristics of the flames, it involves many different physico-chemical processes that are not well understood yet and whose role strongly depends on the particular system. However, some insight in this respect can be gained from experiences with other types of flames.

A preliminary version of the SOFM method was also applied to a commercial oil-fired burner with a nominal capacity of 60 kW. The flames in this case were yellow and highly radiating, with characteristics (physical and visual) very different from those of the blue flames described. In particular, the nature and origin of emission spectra are expected to be deeply different: dominated by black-body radiation from soot particles, with a much smaller contribution due to chemiluminescence of excited radicals. An exercise similar to that reported in Section 5.4 was performed to estimate $NO_x$ concentration from flame images. $NO_x$ emissions varied in the range 53–94 ppm for the flames analysed; the estimation error was within $\pm 5$ ppm in practically all cases, very similar in relative terms to the results shown in Fig. 9.

The probabilistic method was used for the identification of lean-premixed flames. The facility was a dump combustor with an axial swirler (60° vanes) in the mixture injection duct and was operated at a ther-

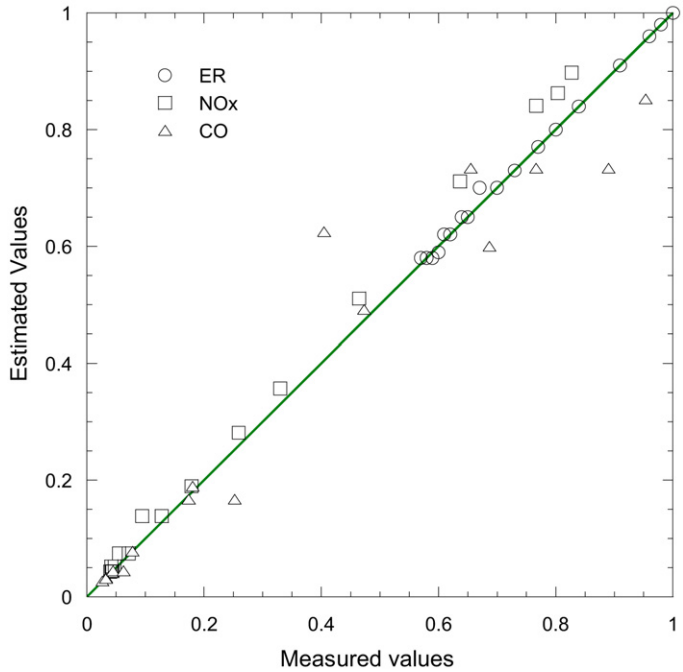


Fig. 16. Comparison between predictions and measurements. $NO_x$ and CO emissions have been normalised by characteristic maximum values (50 and 3000 ppm, respectively).

mal input of 33 kW with natural gas. Equivalence ratio was varied in the range 0.57–1.0, and the objective was to identify the operating conditions using flame images as the only input. The method was applied exactly as described above, initially developed for the diffusion flame, without any adaptation work specific to the new facility. The visual and physical characteristics of the flames were, obviously, very different from those of the air-staged burner. Fig. 16 compares actual and predicted values for equivalence ratio (ER), $NO_x$ and CO emissions, obtained in an identification exercise based exclusively on flame images. Each test flame was assigned the parameters measured in a previous test campaign for the pattern flame yielding the highest probability. Significant deviations resulted only for some CO estimations, although only in cases of relatively high emissions (>1000 ppm) where even experimental repeatability is rather low. In general, the quality of the predictions is considered satisfactory, suggesting that this technique offers also a good potential for the monitoring of premixed flames.

Background radiation was easily removed in all the cases explored by image thresholding. However, this issue might become more critical in combustion chambers with very hot and radiative walls and could require a careful selection of threshold values or removing a part of the incoming radiation by low-pass optical filtering. For the burners studied in this work, image databases included ∼3–4 different values for each of the burner settings (e.g., 49 cases were characterised, including simultaneous variations in 3 independent burner settings). Nevertheless, a wider study would be needed to establish the minimum amount of information needed to apply this type of methods.

## 8. Conclusions and final remarks

The potential of flame imaging as a monitoring tool has been investigated through the application of different processing methods to an image database generated in a model industrial burner. Similarly to other works, parametric analyses confirmed that variations in operating conditions affect significantly some image characteristics. However, much better results were obtained with two other approaches treating the image as a whole. The main differences and novelties with respect to other works in this field are:

- The methods were conceived as general procedures, in contrast with feature-based techniques which always need adaptation to particular situations in terms of feature selection, development of empirical correlations among image features and combustion parameters as well as definition of the most suitable functional forms.
- They have been evaluated in widely different applications. To the authors' knowledge, no previ-

ous work in this field reports attempts to apply and evaluate other methods in more than one situation.

- The methods proposed are suitable to any kind of image. The sensor can be a conventional CCD camera, or any other imaging technique such as planar laser techniques. Moreover, the performance might still improve, as the information generated with advanced diagnostic techniques is usually more intimately related to flame properties and displays stronger gradients and sensitivity to operating conditions than the luminosity in the visible range.

One of the methods exploits the good capabilities of self-organising feature maps for pattern classification (and, therefore, identification). Although these techniques are conceived for qualitative classification, values for the relevant parameters can be readily obtained through the assignation/interpolation procedure proposed. The second method uses an approach similar to those used for speaker recognition. In this case, principal component analysis demonstrated to be an effective data-reduction technique. A probabilistic approach was followed to quantify the degree of similarity of an 'unknown' image with all reference patterns; the representation of patterns by a set of images is thought to be a salient feature of this method, as it provides the robustness required to account for the variability inherent to each combustion state. Both procedures are completely general; none of the operations performed for image analysis (except, perhaps, a part of the pre-processing) was specific to this particular combustion rig. Besides being always a valuable virtue, the generality of the method greatly alleviates one of the main potential drawbacks of empirical approaches, which is the invalidation of the empirical model due to changes in system behaviour. The flame classifiers described can be designed to adapt to new situations (e.g., fuel switching) by updating the reference patterns (even during normal operation) and with little intervention of highly-skilled personnel.

The methods described were also applied to other types of flames, including an oil burner and a lean-premixed combustor, yielding results of quality similar to those presented for the gas diffusion flame. It is important to note that these techniques have been applied, without specific adaptations for the different cases, to widely different flames: oil-/gas-fired, stable/pulsating, diffusion/premixed, fast-mixing/staged, attached/detached. The good results obtained in all cases are thought to strongly support the generalisable nature of the approaches presented. This must be considered, however, an empirical conclusion, as it is not possible to justify on physical grounds the existence of a defined relationship among visible images and flame characteristics for the general case.

This kind of techniques is thought to enable the development of advanced monitoring methods in practical combustion systems, either for the identification of flame state (e.g., control applications, anomaly detection, etc.) or as a virtual sensor for the prediction of relevant parameters such as $NO_x$ emissions. Compared to monitoring based on flue gas analysis, the techniques described offer, besides a much richer diagnostic, the possibility to monitor individual flames in multiburner chambers and a dramatic reduction in response time (evaluation times were 0.9 s for the SOFM and 0.25 s for cepstral analysis), very relevant for control purposes.

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